

# Constitutional Distillation AI Framework for Trust and Personalised AI on Small Language Models

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An AI assistant running on the user’s own device keeps private data off the internet; however, running an assistant on a device requires smaller models, which sacrifice the capacity needed for safety, reasoning, and tool use. This dissertation investigates whether a framework can teach a constitution into a 0.6-billion-parameter model, a small language model from the Qwen3 Family.

This dissertation experiments on five model conditions across three staged runs, which include baseline against a vanilla model, template-based supervised fine-tuning, then constitutional distillation from a frontier-scale teacher, then experimenting on a new Thinker-Executor architecture that splits the dependency across two models. All run on Qwen3–0.6B within a pipeline built around a twenty-five principle constitution, which is judged by a large language-model judge ranking all five answers to each question. The training corpus is synthetic, with its target answers generated by a larger model, as collecting such data is a separate research problem and the goal here is to create a framework.

However, the training data decides the outcome, as seen during evaluation. By replacing a format-only corpus with a constitutional one, over identical weights and recipe, it boosts the head-to-head score by 0.230. That same model does no better on average than an untrained one given a careful prompt and real tools, at 0.589 against 0.583 score. But the average scores hide the area for strengths and limitations. The constitutional improvement also came at a cost. The training that improves compliance cripples the model’s written reasoning, taking the empty-reasoning rate from 1.4% to 91.3% and the mean trace length from 1,309 to 150 characters. This is the small-model form of the *alignment tax* and *safety tax*. Splitting reasoning from execution wins some of it back, taking the empty-trace rate to 36.2% and producing a clarifying question before answering, but it does not match the single model’s compliance and is weakest over long conversations.

The main contribution of this work is a model-agnostic framework for measuring and improving constitutional behaviour at small scale. At this scale, trustworthy behaviour appears to be acquired in layers, not all at once.